CNN: checkpoint

Can a small convolutional neural network (CNN) replace the INT8 LDA in our prosthetic-arm gesture classifier? The models, the fair test set-up, the results as they come in, and the decision

B.Tech Project · FORS-EMG, small CNN against the INT8 LDA on a low-cost chip · Version 11 · 23 October 2026

Version history

Version	Date	What changed
1	22 October 2026	Set-up: models, chip cost on paper, review of the first scripts, rules, test plan, task list
2	22 October 2026	CNN moved onto the documented LDA pipeline (window_test.py); new Section 2.1; review rows 2–4 changed; file list updated
3	22 October 2026	Small CNN designed (3.3); CNN-1 set-up written (9.1); shared code and CNN-1 script ready and checked; task list updated
4	22 October 2026	CNN code made self-contained and saving to Drive; gesture-merging check added to CNN-1
5	23 October 2026	CNN-1 results, insights and decision (9.1); tracker #8–#10
6	23 October 2026	Bars agreed (two +3, small +2); CNN-2 set-up (9.2); CNN-2 script ready and checked
7	23 October 2026	Terms page; CNN-2 results, insights and verdict (9.2); next steps; tracker #11–#14
8	23 October 2026	Next steps agreed: CNN-2b, CNN-2c, one CNN-4 check; CNN-3 and CNN-5 dropped
9	23 October 2026	CNN-2b results (9.3): same setting chosen, verdict final; CNN-2c and CNN-4 running with 3 seeds
10	23 October 2026	CNN-2c results (9.4): more calibration does not change the decision; tracker #15–#16
11	23 October 2026	CNN-4 results (9.5); final verdict and what a CNN would offer (new Section 10); sections 10–12 renumbered 11–13

Where we stand

1. **The question.** Our chip uses an INT8 LDA on 40 hand-made features. A small CNN learns its own features from the raw signal. Is it accurate enough to justify its much larger chip cost, once both are calibrated to the user as in the real product?
2. **Stage: CNN-1 and CNN-2 done (Sections 9.1 and 9.2).** By the rules fixed before the tests, neither CNN is adopted and the LDA stays. Two short checks remain before the branch is closed (Section 8).
3. **CNN-1 result.** Without calibration every CNN beats the LDA: +3.9 points (small), +6.7 (two), +7.1 (sep). INT8 costs nothing. "two" goes forward by the rule fixed beforehand, with "small" kept as the low-cost option. All CNNs are still about 14 points below the calibrated LDA, so CNN-2 decides.
4. **CNN-2 result.** After one calibration trial: LDA 91.9%, "small" 92.2%, "two" 94.3%. "two" is better for all 19 people (+2.4 points, 95% interval +1.9 to +2.9) but below its +3 bar; "small" is level with the LDA (+0.3, bar +2). Calibration closes most of the CNNs' head start: "two" led by 6.7 points before calibration and by 2.4 after.
5. **What the CNN is.** The same last step as the LDA (one layer that turns features into 12 gesture scores), but the 40 fixed features are replaced by 64 learned ones. The learning part filters each signal in time and then mixes the electrodes, four times over (Section 3).
6. **What it costs, on paper.** About 572,000 multiply-adds per decision against the LDA's 520 (≈1,100×), and ≈24 KB of memory against ≈1.5 KB (model plus feature block). At our 1 MHz clock one decision would take ≈0.57 s, but one is needed every 50 ms (Section 4).
7. **How much room there is.** The calibrated LDA already reaches ≈92% held accuracy on FORS-EMG (6 groups). The CNN can gain at most 8 points; the agreed bars are +3 points for "two" and +2 for "small" (Section 6).
8. **CNN-2b confirmed it.** Offered stronger calibration settings, the validation trials chose the same one again; "two" stays at +2.39 [+1.87, +2.90]. The decision is final: **keep the LDA**.
9. **More calibration (CNN-2c) does not change it.** With 2 and 3 trials the CNN's lead is +2.55 and +2.80 points. The CNN with 1 trial (94.3%) already beats the LDA with 3 trials (93.1%): a shorter calibration, recorded for a future, larger chip.
10. **Arm position (CNN-4).** Calibrated in only one arm position, the CNN loses 9.6 points and the LDA 12.1: the CNN is more robust (its lead doubles to +4.9), but both still need a calibration across positions.
11. **Final verdict (Section 10): keep the INT8 LDA.** The CNN is consistently better (more accurate, better before calibration, shorter calibration, more robust to arm position) but by less than the agreed bar, for ≈870 times the arithmetic. Section 10.3 lists what a CNN would offer if a future chip can afford it.

Test	Status	Result in one line
Review of the first CNN scripts	Done (Section 5)	15 changes needed before a fair comparison
CNN-1 · Global model, no calibration	Done (Section 9.1)	INT8 holds; two +6.7 over the LDA before calibration; two (and small) go forward
CNN-2 · Calibration	Done (Section 9.2)	two +2.4 over the calibrated LDA, better for all 19 people, but below its +3 bar; small +0.3 (level). The LDA stays
CNN-2b · Wider calibration settings for two	Done (Section 9.3)	Same setting chosen (0.001, 20 epochs); +2.39 [+1.87, +2.90]; verdict final
CNN-2c · 2 and 3 calibration trials	Done (Section 9.4)	Lead grows only to +2.80 with 3 trials; CNN with 1 trial beats LDA with 3
CNN-3 · Rest and smoothing	Dropped (stopping rule)	Neither CNN passed its bar

Test	Status	Result in one line
CNN-4 · Robustness (position, force, shift, days)	Done: arm-position check (9.5)	CNN more robust: lead +2.4 → +4.9 when calibrated in one position
CNN-5 · NinaPro DB2 benchmark	Dropped (stopping rule)	Neither CNN passed its bar
CNN-6 · Chip cost and hardware impact	Done (Sections 4, 10.2)	≈870× the LDA's multiply-adds for two; ≥9 MHz
Final decision	Keep the LDA (Section 10)	Confirmed by every test

Terms used in this report

Every number in this report is built from the terms below. They are explained once here so that no table needs guessing.

Term	Meaning
Window	A 150 ms piece of the 8 electrode signals (147 samples at FORS-EMG's 985 Hz). The classifier gives one answer per window, and a new window starts every 50 ms
Held window	A window that ends more than 0.5 s after the recording starts, when the person is already holding the gesture. The first 0.5 s, while the hand is still moving into the gesture, is not scored
Gesture accuracy	% of windows whose predicted gesture (one of 12) is correct
Group accuracy	% of windows whose predicted gesture falls in the correct one of the 6 fixed groups. Mixing up two gestures of the same group is not an error
Main measure	Group accuracy on held windows, INT8, mean over the 19 people
Point	One percentage point: going from 92% to 94% is +2 points
Float / INT8	Float: the computer's full precision. INT8: every weight and stored value rounded to an 8-bit integer, as the chip stores them
MAC (multiply-add)	One multiplication added to a running sum; the basic step of both classifiers, used here as the measure of chip work
LDA	The current classifier: 40 hand-made features (5 per electrode) → 12 scores
CNN	A classifier that learns its own features from the samples. Three are tested: "small" (≈49,000 MACs), "two" (≈452,000) and "sep" (≈572,000)
Global (factory) model	A model trained on the other 18 people; it has never seen the new person
Calibration	Adapting the global model to a new user with a short recording of that user
Trial	One recording session: each of the 12 gestures in each of 3 arm positions. Each FORS-EMG person has 5 trials
Calibration / validation / test trial	Used to adapt the model / to choose settings (never to measure) / only to measure accuracy (never for any choice)
Round	One assignment of a person's 5 trials to these roles. Each person has 5 rounds, so every trial is tested once; a person's result is the average of the 5 rounds
Model C	The adopted LDA calibration: re-centre the features on the user's anchor gestures, then blend the factory statistics with the user's
Share	In model C, the fraction of the statistics that comes from the user (0 = factory only, 1 = user only), chosen on the validation trials
Seed	The random starting point of a CNN's training. Each CNN is trained 3 times per person and the 3 results are averaged
Paired difference	A − B computed person by person, then averaged over the 19 people
95% interval	The range in which the true average difference lies with 95% confidence, e.g. +2.4 [+1.9, +2.9]
Up / Down	For how many people A is higher / lower than B
No meaningful difference	The 95% interval lies inside ±1 point (the size of run-to-run noise)
Bar	The gain a CNN must show over the calibrated LDA to be adopted: +3 points for "two", +2 for "small", with the lower end of the 95% interval above +1
Confirm 4	The hand acts only after the same answer comes out 4 times in a row

1. Why look at a CNN?

The LDA only sees what our five features capture: how strong each muscle signal is (MAV, VAR), how fast it changes (WL, SSC) and how often it crosses zero (ZC). Anything else in the signal is thrown away before the classifier sees it, for example the exact shape of the signal over time or timing differences between muscles. A CNN starts from the raw samples and learns which patterns separate the gestures, so in principle it can use information the LDA never sees.

Two results from the LDA work suggest there may be something to gain. On NinaPro DB2 our chip model reaches 53–55% on 17 movements against ≈80% published (Calibration checkpoint, Section 13), and adding three more features moved the accuracy by only 1.4 points, so better hand-made features alone are unlikely to close the gap.

Three things argue against a large gain in our product, and the tests are designed to measure each of them:

- **Calibration already fixes most errors.** One short calibration trial adds ≈20 points to the LDA. Much of what a CNN learns better (differences between people) is exactly what calibration corrects.
- **The group map hides fine differences.** Six of the 12 gestures form group G1, so confusing Index with Peace is not an error for the hand. A CNN's advantage at telling finger shapes apart may disappear after the map.
- **Little data.** FORS-EMG has 19 people. CNNs usually need many more people before they pull ahead of a linear classifier on good features.

2. The baseline the CNN must beat

All numbers come from the Calibration checkpoint and use the adopted LDA: 12 gestures scored, mapped to 6 groups, INT8 with one power-of-two scale per class row.

LDA, FORS-EMG, 19 people	Setting	Group accuracy %	Source
General model, no calibration	200/100, all windows	70.2	Section 5
Model C, 1 calibration trial	200/100, all windows	90.3	Section 5
Model C, calibrated, held	**150/50, no smoothing**	**92.1**	**Section 12.6**
Model C, calibrated, held	150/50, confirm 3	92.5	Section 12.6

- **Held** accuracy ignores the first 0.5 s of each recording, when the hand is still moving into the gesture. It is the figure the window test used, and the one used here.
- The 150/50 number without calibration was not reported. CNN-1 measures it again in the same run as the CNN, so the comparison never mixes runs.
- **FORS-EMG has no Rest recordings.** On FORS-EMG both models therefore score 12 gestures (480 multiply-adds for the LDA). Rest, the 13th class and the strength gate are tested in CNN-3 on the datasets that have Rest (GRABMyo, ContractionIntensity, TMR).
- Chip side: 520 INT8 multiply-adds per decision with Rest, ≈0.7 KB of model numbers, ≈0.8 KB of storage in the feature block, 1 MHz clock.

2.1 The signal path behind these numbers

The documented LDA results come from **fors_emg_lda_baseline.py** with **window_test.py**, not from K_4_LDA_Analysis.py, which the first CNN scripts import. The CNN is therefore moved onto the same path, so both models see identical windows:

Step	Documented LDA (window_test.py)	CNN
Sampling	FORS-EMG's own 985 Hz	Same
Filter	Band-pass + notch, zero-phase (filtfilt), float	Same signal
Window / step	150 ms / 50 ms = 147 / 49 samples	Same windows
Input to the classifier	5 float features per channel, re-centred on the person's anchor gestures (Hand Open, Wrist Flexion, Wrist Extension)	The 147 filtered samples per channel
INT8	Weights: one power-of-two scale per class; inputs: one scale	Section 5, rows 5–6
Held windows	Window ends more than 500 ms after the recording starts	Same

The chip's own signal path (1 kHz, causal filter, 8-bit ADC, integer features) was checked for the LDA separately, in the hardware ladder (fors_emg_hw_bitexact.py, 4 gestures, 200/100). The same check is done for the CNN in CNN-6, so the main comparison is not mixed with the chip's signal path.

3. The CNN models, in simple steps

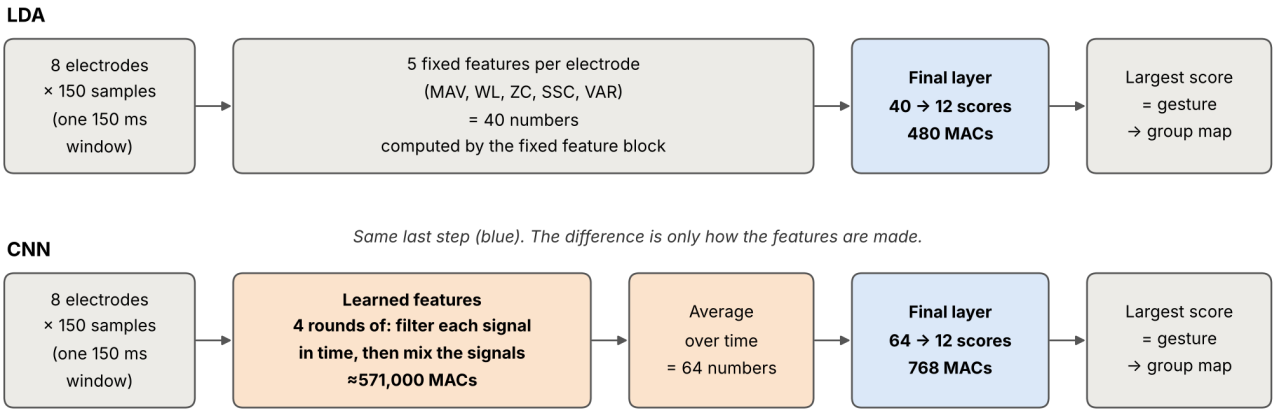


Figure 1. LDA and CNN side by side. Both end with one layer that gives a score per gesture; the largest score wins and the group map turns it into a hand action. Only the feature part differs. One MAC (multiply-add) is one multiplication added to a running sum, the basic operation of both models.

3.1 Model "sep" (option 2), step by step

Input: one 150 ms window, i.e. 8 electrodes × 150 samples. Each step below has two kinds of operation:

- **Filter in time:** each signal is slid past a short learned pattern (5–15 samples long) and the match is recorded at every position. This is the learned version of features like WL or ZC, which also look at how one signal changes over time.
- **Mix:** each new signal is a weighted sum of all signals at the same moment. This is needed because a gesture is a pattern across several muscles, not one muscle alone.
- After each operation, negative values are set to zero (this is what lets the network learn more than a linear rule), and a normalisation step keeps the values in a steady range during training. On the chip the normalisation is merged into the weights, so it costs nothing.

Step	What it does	Output (signals × samples)	MACs	Weights
1	Filter in time: 4 filters of 15 samples per electrode	32 × 150	72,000	480
2	Mix the 32 signals	32 × 150	153,600	1,024
3	Filter in time (7 samples), keep every 2nd result	32 × 75	16,800	224
4	Mix, 32 → 48 signals	48 × 75	115,200	1,536
5	Filter in time (7 samples), keep every 2nd	48 × 38	12,768	336
6	Mix, 48 → 64 signals	64 × 38	116,736	3,072
7	Filter in time (5 samples), keep every 2nd	64 × 19	6,080	320
8	Mix the 64 signals	64 × 19	77,824	4,096
9	Average each signal over time	64 numbers	–	–
10	Final layer: 64 → 12 gesture scores	12 scores	768	768
	Total		**571,776**	**11,856**

Why every second result is kept (steps 3, 5, 7): it halves the length each time (150 → 75 → 38 → 19), which keeps the cost down and lets later filters cover a longer stretch of the window. **Why the signals grow** from 32 to 64: as the length shrinks, more signals keep the amount of information roughly constant.

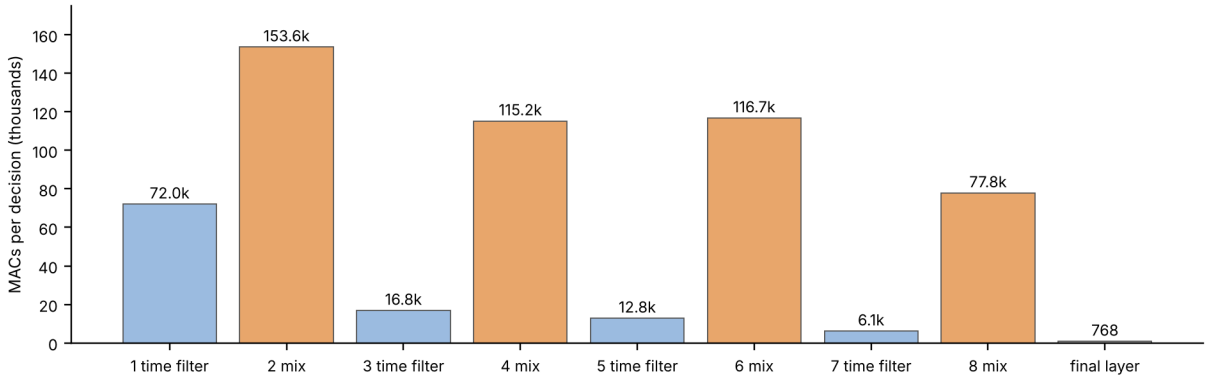


Figure 2. Where the multiply-adds go in model "sep". The four mixing steps (orange) take 81% of the cost; the time filters (blue) only 19%. A cheaper CNN should first use fewer signals, not shorter filters.

3.2 Model "two" (option 5c)

The same building blocks in a different order. The first two steps are done side by side instead of one after the other: one branch only filters in time (each electrode on its own), the other only mixes the electrodes. Their 32 + 32 signals are joined into 64 and passed through two more filter-and-mix rounds. Both branches keep every second result right at the start, which is why this model is ≈20% cheaper (≈452,000 MACs).

3.3 Model "small": a deliberately cheap third model

Both models above cost about 1,000 times the LDA. To show how accuracy changes with cost, rather than two points far from the LDA, CNN-1 adds a model with about 12 times fewer MACs. Following Figure 2, it keeps the structure of "sep" but uses fewer signals, and keeps only every third result in step 1. Counted for a 150-sample window:

Step	What it does	Output (signals × samples)	MACs	Weights
1	Filter in time: 2 filters of 15 samples per electrode, keep every 3rd	16 × 50	12,000	240
2	Mix the 16 signals	16 × 50	12,800	256
3	Filter in time (7 samples), keep every 2nd	16 × 25	2,800	112
4	Mix, 16 → 24 signals	24 × 25	9,600	384
5	Filter in time (5 samples), keep every 2nd	24 × 13	1,560	120
6	Mix, 24 → 32 signals	32 × 13	9,984	768
7	Average each signal over time	32 numbers	–	–
8	Final layer: 32 → 12 gesture scores	12 scores	384	384
	Total		**49,128**	**2,264**

Why every third result in step 1: the band-pass filter already removes everything above 450 Hz, and the learned 15-sample filter smooths further, so little is lost by keeping a third of the results, while every later step becomes three times cheaper.

4. Chip cost, on paper

Counted layer by layer for the deployed setting (150 ms window, 13 classes including Rest, INT8 weights, 32-bit biases). These are paper numbers; CNN-6 will repeat them for the model actually chosen.

Per decision (every 50 ms)	LDA (adopted)	CNN "small"	CNN "sep"	CNN "two"
Multiply-adds (MACs)	520	≈49,000	≈572,000	≈452,000
Model numbers stored	≈0.7 KB	≈2.9 KB	≈13.5 KB	≈13.9 KB
Working memory	40 B features	≈1.6 KB	≈9.6 KB	≈10.8 KB
Signal storage in the feature block	≈0.8 KB running sums	1.2 KB raw samples (+ MAV sums)	same	same
Time per decision at 1 MHz, 1 MAC per clock	0.5 ms	≈49 ms	≈0.57 s	≈0.45 s
Clock needed to finish within 50 ms	≈0.25 MHz (feature block)	≥ ≈1 MHz	≥ ≈12 MHz	≥ ≈9 MHz

- **Working memory** is the largest amount of intermediate signals held at the same time: during one step, its input and its output (for "sep", steps 1–2: 32 × 150 values in and 32 × 150 out, 9,600 bytes).
- **The clock rows are a lower limit.** They assume one multiplier busy on every clock cycle. Reading weights and signals from memory usually adds cycles, so the real clock would be higher, or several multipliers would work in parallel.
- **The feature block changes.** A CNN on raw samples needs the last 150 samples of all 8 electrodes (1.2 KB at 8 bits) instead of five running sums. The MAV sums should stay, because the Rest strength gate uses them (8 additions).
- **Calibration changes too** unless the CNN keeps an LDA as its last layer (option (e) in CNN-2), see Section 7.

5. Review of the first CNN scripts

Two scripts existed when this branch started: **K_4_CNN_LOSO.py** (both CNNs against the LDA, each person left out in turn, with an INT8 simulation) and **K_4_CNN_Calib.py** (fine-tuning the CNNs to one person, S15). Their sound parts are kept: the CNN and LDA see the same windows and folds; early stopping uses other people's data, never the test person's; the normalisation is merged into the weights before INT8; and each electrode's signal is multiplied by a random gain during training, which imitates the contact differences between people. What has to change:

#	Problem found	Why it matters	Change
1	4 gestures, 200 ms windows, decision every 100 ms	Not the current design	12 gestures → 6 groups, 150/50
2	Data, filter and windows come from K_4_LDA_Analysis.py	Not the pipeline behind the documented LDA	Build on fors_emg_lda_baseline.py, as window_test.py does (Section 2.1)
3	CNN gets the float signal, LDA gets 8-bit samples; 8-bit full scale includes the test person	The two models see different signals; small leak	Both use the documented float path; the 8-bit check (CNN-6) takes its scale from training people only
4	LDA reference is K_4's LDA (other features, one INT8 scale)	Not the documented LDA	window_test.py's model C: anchor re-centring, one power-of-two scale per class
5	CNN signal scales between layers are any number	Chip would need a multiplier for each	Powers of two (shifts only), or count the multiplies
6	One weight scale per layer	Small filters round to almost zero	One scale per output signal
7	"Better" whenever the interval excludes 0	Not our ±1-point rule	Use the rules of Section 6
8	One training run per left-out person	Run-to-run noise of 1–2 points	3 runs (seeds), averaged per person
9	Trial 5 of training people used only by the CNN	Different training data	Both train on trials 1–4 (as fors_emg_lda_baseline.py); trial 5 only stops the CNN's training
10	All windows scored, gestures only	Report uses held windows and groups	Held + all windows; gestures + groups; confirm 4
11	Calibration tested on one person (S15)	Cannot support a conclusion	All 19 people, the LDA's 5-round protocol
12	Fine-tuning epochs and learning rate have no validation data	Settings may be tuned on the test	Choose them on the validation trial
13	No INT8 after calibration	Chip runs INT8	INT8 after every calibration
14	The two scripts define different networks	Their numbers cannot be combined	One shared file: cnn_common.py
15	Fine-tuning changes the normalisation and all weights at once	Cannot tell which helped	Separate options (a)–(e) in CNN-2

6. Rules fixed before any test is run

As in the LDA work, the rules are written down before the results exist, so the line cannot move after the numbers are seen.

Rule	Setting
Main measure	Group accuracy (6 groups via the map) on held windows, INT8, per person; mean over all people with a paired 95% interval
Also reported	All windows; 12-gesture accuracy; balanced accuracy; with confirm 4
No meaningful difference	A difference within ±1 point (run-to-run noise is of this size)
INT8 must hold	CNN INT8 − CNN float within ±1 point, else the quantisation is fixed before going on
Choosing between CNNs	Among the CNNs within 1 point of the best, keep the cheapest
Bar for adopting the CNN (agreed 23 October)	**After the same 1-trial calibration, CNN − LDA (model C) on held group accuracy, INT8: at least +3 points for "two" and +2 points for "small", with the lower end of the 95% interval above +1; then no more unwanted movements per minute at rest (CNN-3) and no robustness test worse by more than 1 point (CNN-4)**
Stopping rule	If in CNN-2 even the upper end of a model's interval is below its bar, that model cannot reach it. If neither can, CNN-3 to CNN-5 shrink to one quick check each and the branch closes with a written negative result
Fair set-up	Same people, rounds, trials, 150/50 windows, signal path (Section 2.1) and group map for both models; 3 training runs per CNN and person

Why +3 points for "two": it needs ≈870 times the LDA's multiply-adds, a clock of 9 MHz or more and ≈15 times the memory, so a gain within run-to-run noise (up to 1.5 points on the 19-person mean, CNN-1) does not pay for it. The calibrated LDA is at ≈92%, i.e. ≈8 wrong decisions per 100 held windows; +3 points removes about 4 of every 10 of those. **Why +2 for "small":** it costs ≈100 times the LDA and fits a ≈1 MHz clock, so the chip barely changes and a smaller, still clearly-above-noise gain is worth it. **Why the lower end above +1:** the gain must be certain, not only large on average.

7. Test plan

Test	Question	Data	Compared
CNN-1	Which CNN, and does INT8 hold? Global model only	FORS-EMG, 19 people	"sep", "two", small CNN, LDA; float and INT8
CNN-2	After calibration, does the CNN beat the calibrated LDA? (main test)	FORS-EMG, LDA's 5-round protocol, 0–3 calibration trials	CNN options (a)–(e) against LDA model C
CNN-3	Rest: false activations and unwanted movements per minute; delay with confirm 4	GRABMyo, ContractionIntensity, TMR	CNN and LDA, each with the strength gate
CNN-4	Robustness: arm position, effort, electrode shift, next day	FORS positions, FougnerLP, ContractionIntensity, ElectrodeShift, GRABMyo	Same tests and rules as the LDA
CNN-5	Standard benchmark	NinaPro DB2, 40 people	CNN against our LDA
CNN-6	Final chip cost and hardware impact	Chosen model	MACs, memory, clock, feature block, calibration path

The five ways to calibrate a CNN (CNN-2)

Option	What changes for the user	Numbers sent to the chip
(a) Re-measure the normalisation	Only the average level and spread of each signal, measured on the user's windows; no training	≈800
(b) Final layer only	The last layer is retrained on the user's windows	≈850
(c) Last round + final layer	Steps 7–10 retrained	≈5,400
(d) Everything	All weights retrained	≈12,300
(e) CNN features + LDA	**The 64 averaged CNN outputs feed an LDA instead of the final layer; calibrated with method C, exactly as now**	**≈1,000**

Why option (e) matters most: method C, the stored history and the channel re-alignment all work by keeping a small summary per gesture (≈14 KB). Option (e) keeps that, so every calibration result of the LDA work carries over. Options (b)–(d) retrain with gradient steps, which needs the factory's raw recordings on the phone or PC to avoid forgetting the general model.

8. Task list: what it takes to characterise the CNN

The branch is complete when every row is Done or Dropped, with the reason written down.

#	Task	Output	Status
0	Read the handoff and the first CNN scripts	Sections 1–5	Done
1	Fix the decision rules	Section 6	Done; bars agreed (two +3, small +2)
2	Get the LDA pipeline files (Section 11)	Same windows and LDA as before	Done
3	Shared code cnn_common.py: data and windows as window_test.py, held windows, models, INT8 simulation, LDA reference, scoring	One file used by every test	Done; checked on test data
4	Design the small CNN	Section 3.3	Done
5	CNN-1 script, Colab run, write-up	Section 9.1	Done
6	CNN-2 script, run, write-up	Section 9.2	Done
6b	CNN-2b: re-run calibration (d) for "two" with a wider range of settings (chosen on validation only)	Section 9.3	Done: verdict confirmed
6c	CNN-2c: 2 and 3 calibration trials for "two", way (d), learning rate 0.001, 20 epochs, 3 seeds, and the LDA	Section 9.4	Done
7	CNN-3 Rest and smoothing	Rest results	Dropped (stopping rule): it could only add reasons against adoption, and needs new data
8	CNN-4 robustness tests	One quick check: calibrate in one arm position, test in the others (FORS-EMG, data already cached); 0.001, 20 epochs, 3 seeds	Done (9.5)
9	CNN-5 NinaPro DB2	Benchmark	Dropped (stopping rule), as CNN-3
10	CNN-6 chip cost and hardware impact	Final cost table for the write-up	Done (Section 10.2)
11	Final decision: adopt, reject, or hybrid (e.g. CNN features + LDA)	Section 10	Done: keep the LDA

#	Task	Output	Status
12	If adopted: update the hardware spec and bit-exact model	Updated spec	Not needed (not adopted)

9. Results

Each test gets a section in the same format as the Calibration checkpoint: how we tested it, results, insights, and whether it warrants a change.

9.1 CNN-1 · Global model, no calibration

Questions. Which CNN is worth taking forward? Does INT8 hold for the CNNs? Where does each CNN stand against the LDA when neither has seen the new person? The last one is information only, because the deployed model is always calibrated (CNN-2).

How we test it (script cnn1_global_test.py, shared code cnn_common.py; the two files need no other project file, because the loader, filter, features and INT8 scale of fors_emg_lda_baseline.py are copied into cnn_common.py unchanged and checked to give identical results; caches and results are kept in MyDrive/cnn_branch, so a Colab reset loses nothing):

Item	Setting
People and data	FORS-EMG, all 19 people, 12 gestures, 3 forearm positions, 5 trials; signal path of Section 2.1
Split	Each person is the new user once. Other 18 people: trials 1–4 train both models; trial 5 only stops the CNN's training. New person: all 5 trials are test windows
LDA reference	40 features, scaled with the training people's mean and spread, LDA from class summaries, INT8 with one power-of-two scale per class. No data from the new person
CNN training	Each pass over the data takes 54 windows at random positions from every training recording (a different crop each time); each electrode's signal is multiplied by a random gain; at most 60 passes, stopped when trial 5 stops improving for 10 passes
Runs	3 models × 19 people × 3 seeds = 171 training runs; each saved at once, so a Colab disconnect loses at most one run
CNN INT8	Normalisation merged into the weights; weights INT8 with one power-of-two scale per output signal; every stored signal INT8 with one power-of-two scale; biases INT32
Measures	Main: group accuracy on held windows, INT8. Also: all windows, 12 gestures, balanced, confirm 4
Gesture merging	Both models classify the 12 gestures; the fixed table merges them into the 6 groups (map). The report also gives, per gesture, how often it lands in its own group, for the LDA and each CNN
Rules	Section 6: INT8 holds if INT8 − float has its lower end ≥ −1 point; among the CNNs within 1 point of the best, keep the cheapest

Why the LDA uses no data from the new person here: the CNN in this test has none either. The adopted model C re-centres the features on the user's anchor gestures, which already needs a calibration recording, so it belongs in CNN-2. Measured this way, the LDA gives 70.1% on all windows, matching the Calibration checkpoint's 70.2%.

Checks before the real run (on made-up signals, so only the code is tested): merging the normalisation into the weights changes the outputs by less than 0.0001%; the INT8 model agrees with the float model on 98% of windows and every scale is a power of two; an interrupted run continues where it stopped; the costs printed by the script match Sections 3 and 4.

Machines. 165 runs on Colab's T4 GPU, the last 6 (S19, "two" and "sep") on a laptop RTX 3050, all trained in float32. 171 runs, about 6 hours of GPU time.

Results

Model	MACs	Train	Val	Test float	Test INT8 (main)	INT8, confirm 4	12 gestures
LDA (no user data)	480	–	–	71.3	71.5 [66.3, 76.6]	72.6	45.7
CNN small	48,632	84.1	82.3	75.5	75.3 [71.0, 79.7]	78.1	51.2
CNN two	**444,608**	**91.3**	**89.1**	**78.1**	**78.1 [73.7, 82.5]**	**80.5**	**54.7**
CNN sep	562,096	91.6	89.2	78.7	78.6 [74.0, 83.2]	80.8	55.5

Group accuracy %, mean over 19 people [95% interval]; each person's value is the mean of 3 seeds. Train / Val = the training people's own windows and their trial 5. 12 gestures = all windows, before the group map.

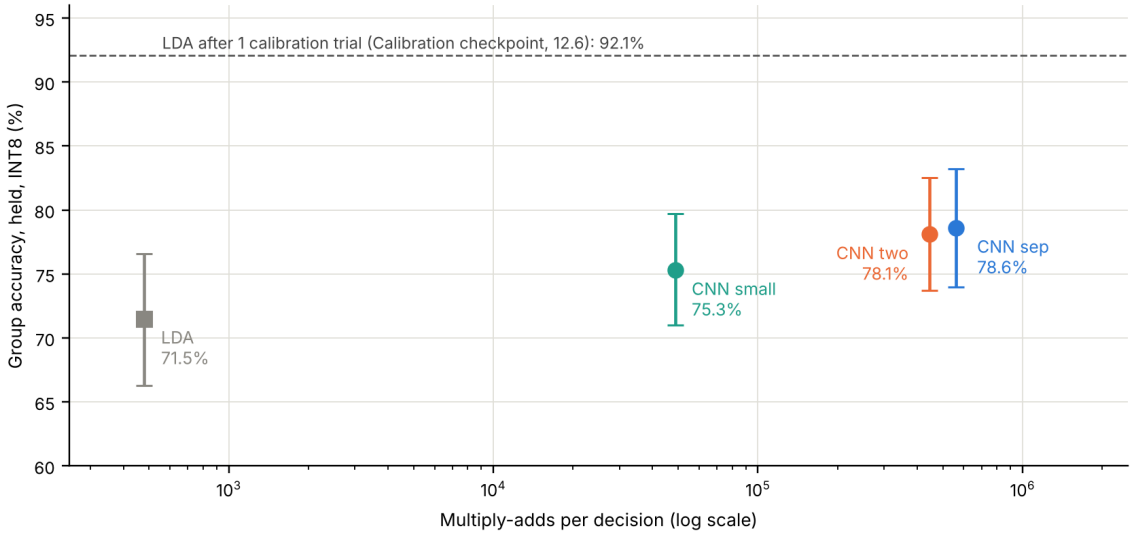


Figure 3. Accuracy against chip cost, no calibration. Every CNN beats the LDA, but all stay well below the LDA after one calibration trial (dashed line). Going from small to two buys 2.8 points for 9× the cost; from two to sep, nothing measurable.

Comparison (main measure, INT8)	Δ points [95% interval]	Up / Down	Verdict
small INT8 − small float	−0.14 [−0.27, −0.01]	4 / 14	No meaningful difference
two INT8 − two float	−0.01 [−0.14, +0.11]	8 / 11	No meaningful difference
sep INT8 − sep float	−0.13 [−0.31, +0.06]	6 / 13	No meaningful difference
small − LDA	+3.86 [+1.66, +6.06]	16 / 3	Better
two − LDA	+6.65 [+3.56, +9.74]	17 / 2	Better
sep − LDA	+7.14 [+4.08, +10.19]	18 / 1	Better
small − two	−2.80 [−4.31, −1.28]	4 / 15	Worse
two − sep	−0.49 [−1.10, +0.13]	7 / 12	Inconclusive (interval just outside ±1)

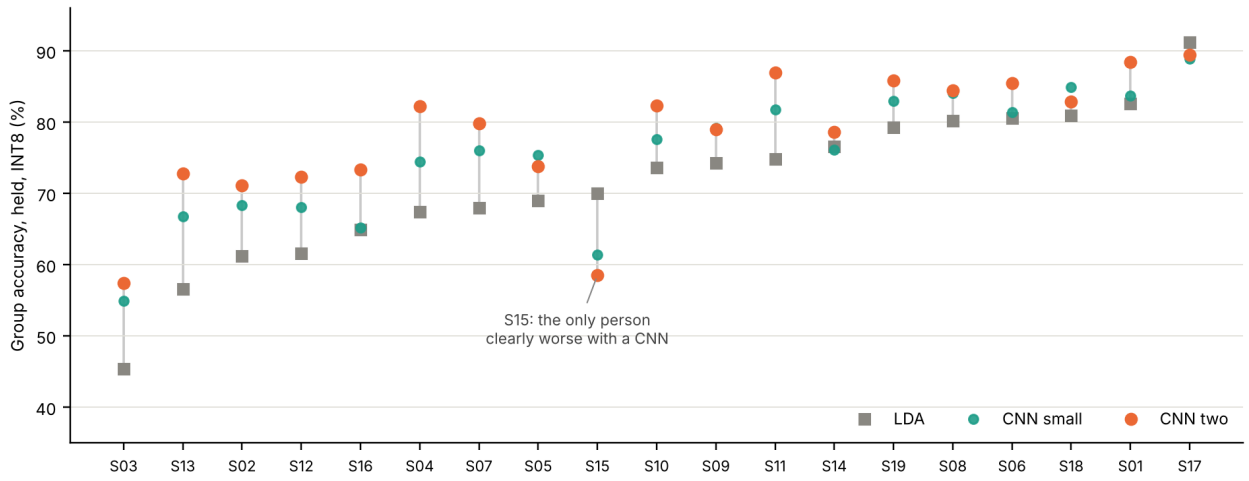


Figure 4. Per person, sorted by the LDA. The CNNs gain most where the LDA is weakest (S03, S13, S04). S15 is the one clear exception.

Gesture merging check

Gesture	Group	LDA	small	two	sep
Hand Close	G1	92.6	93.1	94.0	93.9
Hand Open	G2	65.4	69.6	71.7	72.9
Index	G1	80.1	79.6	82.6	82.5
Index Little	G1	**81.7**	68.2	73.7	74.9
Peace	G1	77.3	81.4	85.7	86.6
Radial Deviation	G3	60.2	68.8	73.4	73.3
Right Angle	G1	75.8	78.6	82.1	80.4
Thumb Little	G4	44.1	54.7	56.2	57.8
Thumb Up	G1	73.8	80.5	85.1	86.8
Ulnar Deviation	G5	67.5	73.5	76.5	77.0
Wrist Extension	G2	79.1	83.6	84.0	84.0
Wrist Flexion	G6	60.2	72.4	72.4	73.1

% of each gesture's held windows that land in its own group (INT8, mean over people).

Insights

- **INT8 holds for every CNN.** The loss is 0.01–0.14 points, far inside ±1. One power-of-two scale per output signal is enough, so the chip needs only shifts between layers and the quantisation needs no change.
- **Before calibration, the CNNs are clearly better than the LDA**: +6.7 points for "two", better for 17 of 19 people. The gain is largest for the people the LDA handles worst (S03 +12, S13 +16, S04 +15). This fits the idea that learned features vary less from person to person than our 5 hand-made ones.
- **But the CNNs are still far from the calibrated LDA** (78% against 92%). They reach 89–92% on people they were trained on and 78% on a new person; this 11–13 point gap between people is exactly what calibration closes for the LDA. Whether a CNN keeps an advantage after calibration is the open question for CNN-2.
- **Bigger stops paying off early.** small → two: +2.8 points for 9× the multiply-adds. two → sep: +0.5, within noise, for 1.3× more. Accuracy levels off around 450,000 multiply-adds per decision.

- **Part of the CNN's advantage is hidden by the group map,** as expected in Section 1: on the 12 gestures "sep" gains 9.8 points over the LDA, on the 6 groups 7.1.
- **The groups still suit the CNNs.** 11 of the 12 gestures land in their own group at least as often as with the LDA, most clearly Wrist Flexion, Thumb Little and Radial Deviation (+12 to +14). The one exception is Index Little (81.7% → 68–75%): worth watching, not enough to re-derive the groups.
- **One person goes the other way:** S15 drops from 70% to 57–61% with every CNN. S15 is also the person the first calibration script was built around; CNN-2 shows whether calibration recovers it.
- **Run-to-run noise is real.** A different seed moves one person by 2–3.5 points on average (up to 14 for "small"), and the 19-person mean by up to 1.5. One seed would not have been enough to separate "two" from "sep".
- **The LDA reference matches the earlier work:** 70.1% on all windows here against 70.2% in the Calibration checkpoint, so the CNN is compared with the same LDA.
- **Smoothing helps the CNNs a little more:** confirm 4 adds 2.4–2.8 points to the CNNs and 1.1 to the LDA.

Does it warrant a change?

Question	Answer
Does INT8 hold?	Yes, for all three CNNs (rule: lower end ≥ −1 point). No quantisation change
Which CNN goes forward?	**"two"**: "sep" is within 1 point of it and "two" needs 21% fewer multiply-adds (rule fixed in Section 6)
Also carry "small"?	Yes (agreed). It is 9× cheaper than "two" and fits a ≈1 MHz clock; if calibration narrows the gap between CNNs, it could be the better trade. Adding it to CNN-2 costs little, because CNN-2 starts from the saved CNN-1 models
CNN against LDA?	Better before calibration (information only). The decision waits for CNN-2

9.2 CNN-2 · Calibration (the main comparison)

Question. After the same short calibration, is a CNN clearly better than the calibrated LDA? This is the test the bars of Section 6 are applied to.

How we test it (script cnn2_calibration_test.py, same shared code):

Item	Setting
Models	"two" and "small", starting from the 3 × 19 global models saved by CNN-1 for each; nothing is trained from scratch
Rounds	For each person, 5 rounds: test = trial k, validation = trial k+1, calibration = trial k+2 (2 trials: k+2, k+3; 3 trials: k+2 to k+4). Every trial is test, validation and calibration once
LDA reference	Model C as adopted (anchor re-centring, blend with the general summaries, share chosen on validation, INT8 per-class scales), on exactly the same rounds and trials
CNN calibration	The five options of Section 7: (a) re-measure the normalisation, (b) final layer, (c) last round + final layer, (d) all weights, (e) CNN features + LDA with method C
Choices	Learning rate and length for (b)–(d), and the share for (e), are chosen on the validation trials, one value for everyone; the CNN's calibration method is the option with the best validation accuracy. Test windows are never used for a choice
INT8	Every calibrated model is INT8. For (a)–(d) the signal scales are re-measured on the user's calibration windows; for (e) the CNN keeps its factory scales
Runs	2 models × 19 people × 3 seeds × 5 rounds; first with 1 calibration trial (the decision), then 2 and 3 trials for the chosen method
Decision	Section 6: "two" ≥ +3, "small" ≥ +2 points over model C, lower end of the interval above +1

One change from the LDA study: the LDA study tried all 3 possible calibration trials in each round and averaged them. Here each round uses one, chosen so that every trial calibrates once. This cuts the run time by 3, and the LDA is recalculated on the same choice, so the comparison stays paired and fair.

Checks before the real run (made-up signals, code only): every option runs, the choices are made on validation only, the decision follows the bars, an interrupted run continues, and the LDA reference is recomputed on the same rounds. Expected run time on a T4 or the laptop: about 4–6 hours for 1 calibration trial (the script's quick mode measures it first).

CNN-2 results, step 1: before and after calibration

All numbers below use the main measure (group accuracy on held windows, INT8) with **one calibration trial**. A person's value is the average over the 5 rounds (and, for the CNNs, the 3 seeds); the table shows the mean over the 19 people with its 95% interval.

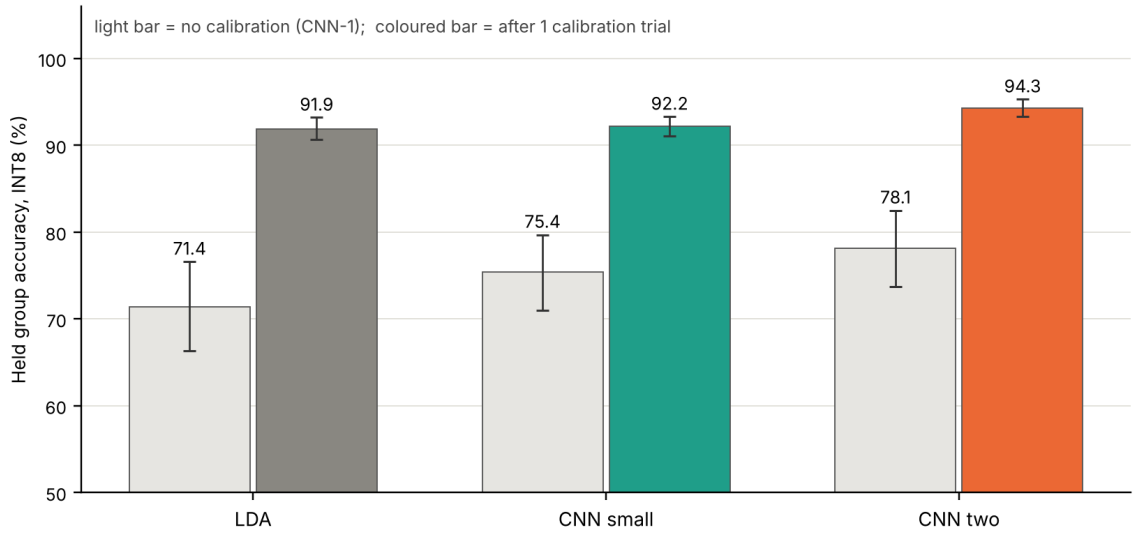


Figure 5. Accuracy before calibration (CNN-1) and after one calibration trial (CNN-2). Calibration lifts every model by 16–20 points, and the models end up much closer together than they started.

Model	No calibration	After 1 trial	Gain from calibration
LDA (model C)	71.4	91.9 [90.7, 93.2]	+20.5
CNN small (calibration d)	75.4	92.2 [91.1, 93.3]	+16.9
CNN two (calibration d)	78.1	94.3 [93.3, 95.3]	+16.2

- **The LDA reproduces the earlier work:** 91.9% here against 92.1% in the Calibration checkpoint (Section 12.6), so the CNNs are compared with the same LDA.
- **Calibration closes most of the CNNs' head start.** Before calibration "two" led the LDA by 6.7 points; after it, by 2.4. The CNN mainly handles differences between people better, and those are exactly what calibration corrects.
- **The LDA's best share was 1.0**, i.e. after re-centring, the calibrated LDA uses only the user's own trial. With one full trial of all 12 gestures, the factory statistics added nothing measurable on the validation trials.

CNN-2 results, step 2: which way of calibrating a CNN works

Each CNN was calibrated in five ways, each starting from the same global model. The settings of each way, and the choice of way, were made on the validation trials only.

Way of calibrating	What changes	small − LDA	two − LDA
(a) normalisation only	Only the average level and spread of each internal signal are re-measured on the user; no training	−12.7	−9.6
(b) final layer only	The last layer (features → 12 scores) is retrained	−3.7	−0.9
(c) last round + final layer	The last filter-and-mix round is retrained too	−1.9	+1.5
(d) all weights	**Every weight is retrained (chosen for both CNNs)**	**+0.3**	**+2.4**
(e) CNN features + LDA	The CNN's 64 (or 32) learned features feed an LDA that is calibrated with model C, exactly like the current design	−3.8	−0.5

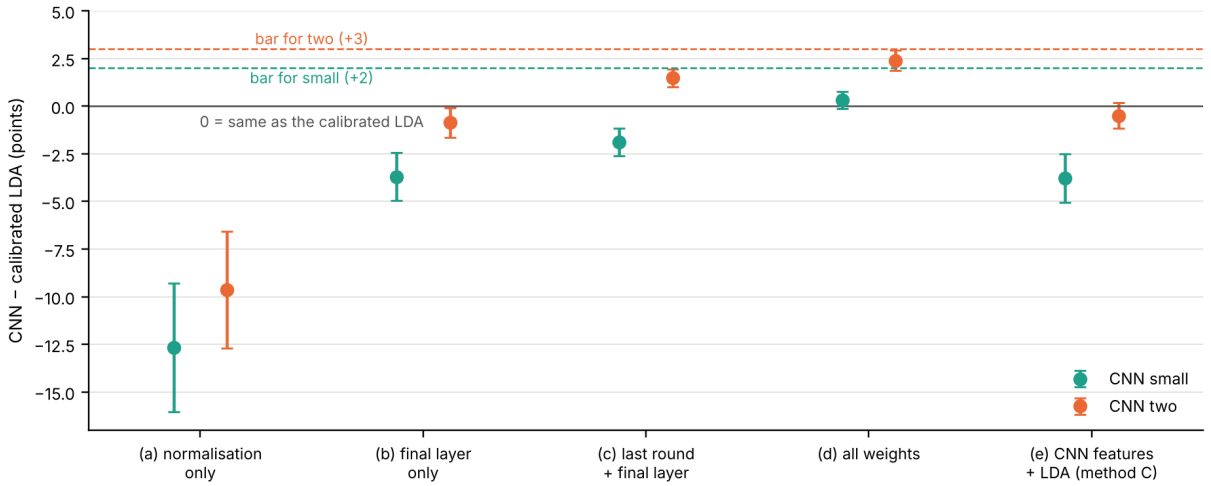


Figure 6. Each way of calibrating, as the difference from the calibrated LDA (points, with 95% intervals). Only retraining all weights (d) brings "two" near its bar, and even then the whole interval stays below it.

- **The more of the network is adapted, the better**: (a) < (b) < (c) < (d) for both CNNs.
- **Re-measuring the normalisation alone (a) is not enough** (+4 points over no calibration). This is the same lesson as the LDA's "re-centre only" (method A): the weights themselves must adapt.
- **The option that keeps the LDA's calibration machinery (e) is only level with the LDA** for "two" (−0.5) and worse for "small" (−3.8). So the CNN's lead needs every weight retrained on the phone or PC at each calibration, instead of the LDA's simple summary blend.
- **Calibration never made anyone worse** than their global model (0 people below by more than 1 point, every way).

CNN-2 results, step 3: the decision

Model	Bar	CNN − LDA after calibration	Up / Down	Verdict (rule fixed before)
CNN two, way (d)	+3	+2.39 [+1.86, +2.93]	19 / 0	Cannot reach the bar: even the upper end (+2.93) is below +3
CNN small, way (d)	+2	+0.32 [−0.13, +0.76]	13 / 6	Cannot reach the bar; no meaningful difference from the LDA

What +2.4 points means in practice. The calibrated LDA gets about 8 of every 100 held windows wrong (8.1%); "two" gets about 6 wrong (5.7%). So "two" removes about 3 in 10 of the LDA's mistakes. With confirm 4 the gap is slightly smaller (92.2% against 94.4%, +2.2). "small"

removes almost none (7.8% wrong).

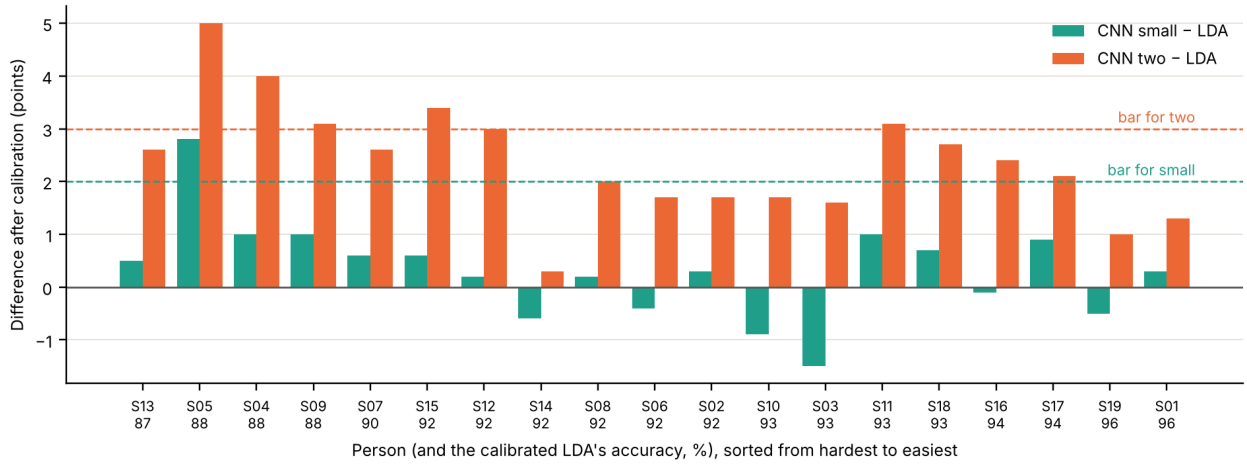


Figure 7. The difference from the calibrated LDA for every person, hardest person on the left. "two" is ahead for all 19 people and gains most where the LDA is weakest (S05 +5.0, S04 +4.0); "small" stays within about ±1.5 points except S05.

Conclusion of CNN-2, by the rules fixed before the test: neither CNN is adopted, and the LDA stays. "two" is genuinely and consistently better, but by less than its bar asks for a model that needs ≈870 times the LDA's multiply-adds and a 9 MHz or faster clock. "small" costs ≈100 times the LDA for no measurable gain.

CNN-2: other observations

- **INT8 still holds** after calibration (INT8 − float: −0.05 points for "two", −0.11 for "small").
- **S15 is recovered.** It was the one person the global CNNs handled badly (58.5% for "two"); after calibration it reaches 95.2%, more than the LDA's 91.8%.
- **The CNN's lead is twice as large on the 12 gestures** (two 83.5% against the LDA's 78.1%, +5.4) as on the 6 groups (+2.4). The group map hides half of it. If the final design ends up with many more groups (the handoff allows 2 to 12), this comparison should be repeated.
- **The hardest people stay the hardest** for every model (S13, S05, S04, S09 at 88–92%), so their difficulty probably lies in their signals rather than in the classifier.

CNN-2: limits of this result

- **Calibration settings at the edge of the range tried.** For (c) and (d) the validation trials picked the largest learning rate and the most epochs tried (0.001, 20 epochs); for (b) also the largest values. A wider range might have helped the CNN. **Checked in CNN-2b (Section 9.3): it does not.**
- **Way (d) is the hardest to put into the product.** Every calibration retrains about 12,000 weights on the phone or PC from a fresh copy of the factory model, and the LDA's daily recalibration with stored history (summaries per gesture) would need a new design.
- 19 people, one session each; one calibration trial per round (Section 9.2, set-up).

CNN-2: does it warrant a change?

Question	Answer
Adopt a CNN?	**No**, by the rule fixed before the test: "two" +2.4 [+1.9, +2.9] is below its +3 bar; "small" +0.3 is level with the LDA
Change to the chip design?	None. The LDA, its features and its calibration stay as they are
Best way to calibrate a CNN, if one is ever used	Retrain all weights (d), with learning rate 0.001 for 20 epochs on one trial
What could change the verdict later	A final design with many more groups; a much larger factory set on our own hardware; cheaper CNN hardware

9.3 CNN-2b · A wider range of calibration settings for "two"

Why this check. In CNN-2, the validation trials chose the strongest calibration setting that was tried for way (d): learning rate 0.001 and 20 epochs. When the best choice sits at the edge of the range, a stronger setting outside the range might be better still, and the CNN's result could be understated.

How we tested it. Exactly as CNN-2 (same people, rounds, trials, seeds and LDA), but "two" with way (d) was now offered four settings: learning rate 0.001 or 0.003, each with 20 or 40 epochs. The choice was again made on the validation trials only.

Result	Value
Setting chosen on validation	Learning rate 0.001, 20 epochs: the same as in CNN-2
CNN two − LDA (model C), main measure	+2.39 [+1.87, +2.90], better for 19 of 19 people
Verdict (bar +3)	Cannot reach the bar: the upper end (+2.90) is below +3
INT8 − float	−0.06 points: INT8 holds

- **The stronger settings were not better on validation.** Counting both runs, the learning rates 0.0003, 0.001 and 0.003 and 5, 20 and 40 epochs were tried, and 0.001 with 20 epochs sits in the middle of that range. So the best setting has been found, and the CNN-2 result was not held back by the range tried.
- **The result repeats CNN-2 almost exactly** (+2.39 in both, with individual people differing by at most 0.3 points from small run-to-run differences on the GPU).
- **The verdict is final:** "two" is consistently better than the LDA by about 2.4 points, but does not reach its +3 bar. The LDA stays.
- **Steps 6c and CNN-4 use this setting** (0.001, 20 epochs).

9.4 CNN-2c · More calibration data: 2 and 3 trials

Why this check. The decision is made with one calibration trial, because that is what the product asks of a user (about 5 minutes: each of the 12 gestures in each of the 3 arm positions). This check records what happens when the user records more: does the CNN's lead grow or shrink? It is information only and cannot change the decision.

How we tested it. As CNN-2: same people, rounds, seeds and LDA (model C, share chosen on validation for each number of trials). Calibration used 2 trials (k+2, k+3) or 3 trials (k+2 to k+4). "two" used way (d) with the setting chosen in CNN-2 and CNN-2b (learning rate 0.001, 20 epochs).

Calibration recorded by the user	LDA (model C)	CNN two, way (d)	CNN − LDA [95% interval]	Up / Down
1 trial (≈5 min) — CNN-2	91.9	94.3	+2.39 [+1.87, +2.90]	19 / 0
2 trials (≈10 min)	92.7	95.2	+2.55 [+2.04, +3.06]	19 / 0
3 trials (≈15 min)	93.1	95.9	+2.80 [+2.32, +3.27]	19 / 0

Held group accuracy, INT8, %, mean over 19 people. With no calibration the two models are at 71.4% (LDA) and 78.1% (CNN two).

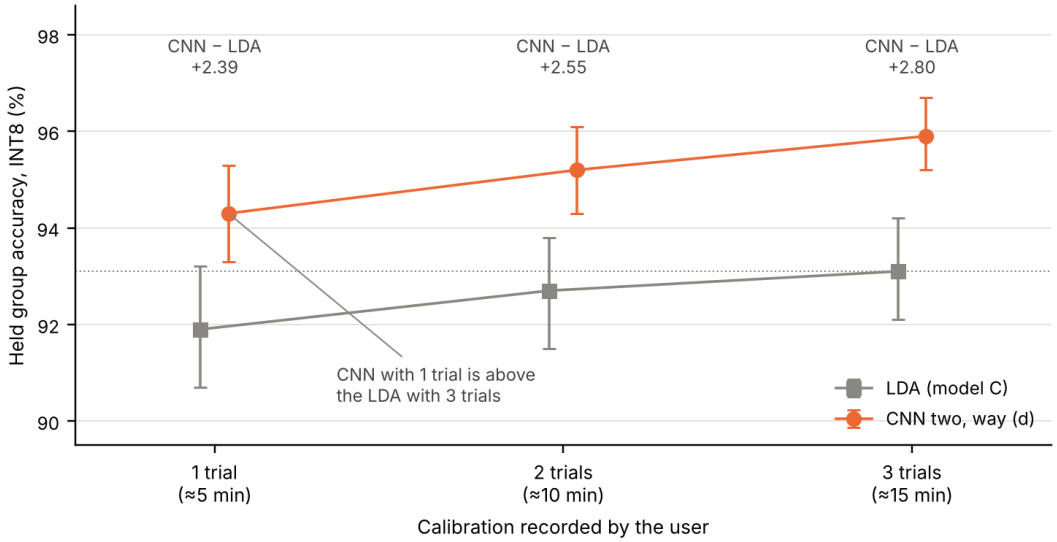


Figure 8. Accuracy against the amount of calibration the user records. Both models improve slowly after the first trial; the CNN's lead grows a little (+2.4 to +2.8 points).

- **The first trial does almost all the work.** One trial lifts the LDA by 20.5 points; two more trials add only 1.2. The CNN behaves the same way (+16.2, then +1.6). A short calibration is enough for both models, which confirms the 1-trial design.
- **The CNN's lead grows only slightly with more data**: +2.39, +2.55, +2.80 points, always for all 19 people. Even with 3 trials (three times the user's effort) the average lead stays below 3 points, so the decision would be the same at any of these lengths.
- **A different way to see the CNN's advantage: shorter calibration.** The CNN with 1 trial (94.3%) is already above the LDA with 3 trials (93.1%). In other words, a CNN could give the user better accuracy for a third of the calibration time. This is a real benefit for the user, but it does not pay for ≈870 times the arithmetic on our chip; it is recorded for a future, larger chip.
- **The rest repeats CNN-2:** INT8 still costs nothing (−0.06 points), the LDA's best share is again 1.0, and nobody gets worse through calibration. With 3 trials the gap is +6.6 points on the 12 gestures (80.6% against 87.2%) and +2.7 with confirm 4.

Question	Answer
Does more calibration change the decision?	No. The CNN's lead grows from +2.4 to +2.8 points and stays below the +3 bar
Should the product calibrate longer?	No. Beyond one trial both models gain little (about +1 point for two more trials, i.e. 10 more minutes)
What is recorded for the future?	A CNN reaches with one trial what the LDA does not reach with three: worth reconsidering if a future chip can afford it

9.5 CNN-4 (quick check) · Calibrating in only one arm position

Why this check. The Calibration checkpoint (Section 9) showed that an LDA calibrated in only one arm position loses about 13 points when the arm moves to another position. That is why the adopted calibration covers several positions. This check asks: is the CNN more or less sensitive to arm position than the LDA? It is the one robustness test kept after CNN-2 (CNN-3 and CNN-5 were dropped by the stopping rule).

How we tested it. FORS-EMG records every trial in 3 forearm positions (Rest, Supination, Pronation). Same rounds as CNN-2, and each model was calibrated in two ways: with the whole calibration trial (all 3 positions), or with only one position of it. Both were then tested on the same windows: the test trial in the **two other** positions. This was repeated with each position as the calibration position and averaged. LDA: model C, share chosen on validation; CNN "two": way (d), learning rate 0.001, 20 epochs, 3 seeds.

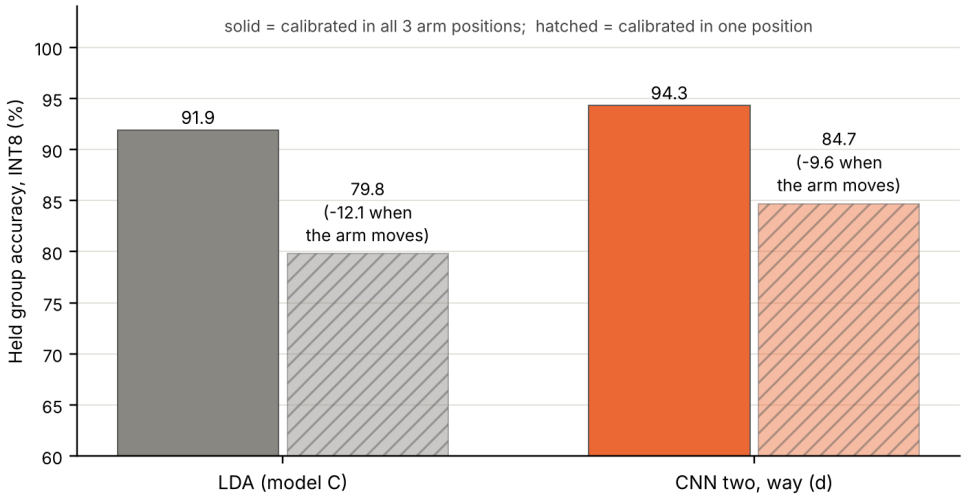


Figure 9. Both models lose accuracy when calibrated in one arm position and used in another; the CNN loses less (−9.6 against −12.1 points).

Comparison	Δ points [95% interval]	Up / Down
LDA: one position − all positions	−12.11 [−14.49, −9.73]	
CNN two: one position − all positions	−9.64 [−12.08, −7.19]	
CNN − LDA, calibrated in all positions	+2.39 [+1.87, +2.90]	19 / 0
CNN − LDA, calibrated in one position	+4.86 [+3.61, +6.10]	18 / 1
Change of the CNN's lead	+2.47 [+1.31, +3.63]	Better (CNN more robust)

- **The LDA result repeats the Calibration checkpoint:** −12.1 points here against about −13 there, so the test reproduces the earlier finding.
- **The CNN is clearly more robust to arm position.** It loses 2.5 points less, and its lead over the LDA doubles (from +2.4 to +4.9 points, better for 18 of 19 people). A likely reason: the CNN learns patterns that change less when the forearm rotates.
- **But both models still lose about 10–12 points.** Calibrating in several arm positions stays essential for either model; the CNN reduces the problem, it does not remove it. The adopted design (calibration across positions) avoids this situation, so the decision at the main measure stands.
- **A useful result for the LDA itself:** when the calibration covers only one position, the best share drops from 1.0 to 0.5, i.e. the factory statistics then carry half the weight. The blend in model C is worth keeping: it protects the user whenever the calibration does not cover everything.
- **The CNN-4 condition of Section 6 is met** (the CNN is not worse than the LDA here; it is better). This cannot change the decision, which was already made at CNN-2.

10. Final verdict

Keep the INT8 LDA. The CNN "two" is more accurate in every test, for almost every person, but by less than the bar fixed before the tests, while it needs about 870 times the arithmetic of the LDA, a much faster clock and a much heavier calibration. The cheaper CNN "small" fits the chip almost as it is, but after calibration it is no better than the LDA.

10.1 The evidence in one table

Test	Question	Result (CNN two − LDA, held group accuracy, INT8)
CNN-1	New user, no calibration	+6.7 points; INT8 costs nothing; "two" chosen
CNN-2	**After 1 calibration trial (the decision)**	**+2.39 [+1.87, +2.90], 19 of 19 people; bar +3 not reached. "small": +0.32, level**
CNN-2b	Wider calibration settings	Same setting chosen; +2.39 again: verdict final
CNN-2c	2 and 3 calibration trials	+2.55 and +2.80: still below 3
CNN-4	Calibrated in one arm position	+4.86: the CNN is more robust, but both lose 10–12 points

10.2 What the CNN would cost on our chip

Per decision (every 50 ms)	LDA (adopted)	CNN two	CNN small
Multiply-adds	520	≈452,000 (≈870×)	≈49,000 (≈95×)
Clock needed, one multiplier	≈0.25 MHz (feature block)	≥ 9 MHz	≈1 MHz
Memory (model + working + signal)	≈1.5 KB	≈26 KB	≈5.7 KB
Calibration on the phone or PC	Blend of summaries per gesture (closed form)	Retrain ≈12,000 weights by gradient steps	Retrain ≈2,300 weights
Accuracy after 1 calibration trial	91.9%	94.3%	92.2%

Memory for "two": ≈13.9 KB of model numbers + ≈10.8 KB working memory + 1.2 KB sample buffer; for "small": ≈2.9 KB + ≈1.6 KB + 1.2 KB (Section 4).

10.3 What a CNN would give us, if the hardware could afford it

Measured in this branch (FORS-EMG, 19 people):

- **Higher accuracy after calibration**: about 3 in 10 of the LDA's mistakes removed, for every person, and most for the hardest users (+4 to +5 points).
- **A much better start for a new user** (+6.7 points before any calibration), so the hand works better on day one and if a calibration is poor or skipped.
- **Shorter calibration**: with 1 trial (≈5 minutes) the CNN already beats the LDA with 3 trials (≈15 minutes).
- **More robust when the arm moves**: its lead doubles when the calibration missed some arm positions.
- **Finer gestures**: its lead on the 12 separate gestures (+5.4) is twice that on the 6 groups, so a CNN could support more distinct commands with fewer merged gestures.
- **No hand-made features**: the same network design works directly on the samples, without choosing, tuning or checking features bit-exactly for each new chip.

Expected from experience with CNNs, but **not tested here**:

- **Easier to move to new hardware or users**: a new electrode layout, sample rate or amputee group needs retraining, not new features.

- **Gains more from more people**: CNNs usually keep improving as the factory set grows (the planned 50–100 people on our own hardware), while a linear model levels off sooner.
- **More from the same signal**: timing between muscles and the shape of the onset are available to a CNN, which could help faster decisions or estimating effort for proportional speed from the same network.
- **Robustness by training**: effects such as gain changes or electrode shifts can be trained in by changing the training data (as the random per-electrode gain already does) instead of redesigning features.

The price, besides chip area and power: a 9 MHz or faster clock, ≈17 times the memory, gradient training on the phone or PC at every calibration, a new design for the daily recalibration with stored history, and a model that is harder to inspect than 40 named features.

10.4 What this branch confirms for the LDA design

- **One calibration trial is the right length**: two more trials add only about 1 point for either model.
- **Calibrate in several arm positions**: calibrating in one position costs 10–12 points for either model.
- **Keep the blend of model C**: with full calibration the user's data dominates (share 1.0), but with partial calibration the factory statistics carry half the weight.
- **The 6 groups suit both models**, and shift-only INT8 (power-of-two scales) works for networks as well as for the LDA.

10.5 When to reopen the question

- The chip budget allows about 9 million multiply-adds per second (for example a microcontroller with a hardware multiplier running at tens of MHz, if the power budget allows it).
- The final design needs many more groups or commands, where the CNN's lead is larger.
- The factory set on our own hardware (50–100 people) shows the CNN gaining more than the LDA as people are added.
- A shorter calibration, or robustness to arm position, becomes a product requirement.

11. Open questions

Item	Why	Status
LDA pipeline files	window_test.py, rest_test.py, fors_emg_hw_bitexact.py, fors_emg_lda_baseline.py, emg_gesture_merging.py; the parts CNN-1 uses are copied into cnn_common.py	Received
K_4_LDA_Analysis.py	The 4-gesture integer-feature LDA; not the documented pipeline	Not used
How "General model only" (Section 5 of the Calibration checkpoint) was computed	CNN-1's LDA with no user data gives 70.1% on all windows, matching the documented 70.2%	Closed
Carry "small" into CNN-2 next to "two"	Not required by the rule; cheap to add (Section 9.1)	Agreed
Bar for adopting the CNN	+3 points for "two", +2 for "small" (Section 6)	Agreed
Run CNN-2b (wider calibration settings for "two")?	The chosen settings were at the edge of the range tried; this decides whether the verdict is final	Done (9.3)
Drop CNN-3 and CNN-5; one quick arm-position check for CNN-4?	Neither CNN passed its bar; these tests could only add reasons against adoption, and need new data	Agreed

12. Improvement tracker

Every test either confirms the design or changes it. This list records each insight and what it changes.

#	Insight (source)	Change	Status
1	The mixing steps are 81% of the CNN's cost (layer count, Section 3)	Small CNN uses fewer signals first	Done: small is 9× cheaper than two for −2.8 points (CNN-1)
2	Retraining CNN weights needs the factory's raw data on the host (Section 7)	Test CNN features + LDA calibrated with method C	Checked (CNN-2): only level with the LDA; the CNN's lead needs all weights retrained
3	Raw-sample CNN needs a 1.2 KB sample buffer; the Rest gate still needs MAV sums (Section 4)	Keep MAV sums next to the buffer	Closed: not needed (CNN not adopted)
4	FORS-EMG has no Rest recordings (Section 2)	FORS tests score 12 gestures; Rest moves to CNN-3	Adopted
5	The documented LDA used 985 Hz, a zero-phase float filter and float features; the chip path was tested only in the hardware ladder (Section 2.1)	CNN compared on the same float path; chip path checked for both in CNN-6	Adopted
6	Model "small" needs ≈49 ms per decision at 1 MHz with one multiplier (Section 4)	If it is accurate enough, the chip clock barely changes	Closed: level with the LDA after calibration (CNN-2)
7	The 6 groups were made from the LDA's confusions; a CNN may confuse other gestures	CNN-1 reports, per gesture, how often it lands in its own group; re-derive the groups only if a CNN differs clearly	Checked: groups hold; watch Index Little
8	INT8 with power-of-two scales costs ≤0.14 points for every CNN (CNN-1)	No change to the quantisation	Adopted
9	S15 is the only person clearly worse with every CNN (−9 to −13 points, CNN-1)	Check whether calibration recovers it	Closed: recovered (two 95.2%, CNN-2)

#	Insight (source)	Change	Status
10	One seed moves a person by 2–3.5 points on average (up to 14) and the 19-person mean by up to 1.5 (CNN-1)	Keep 3 seeds in every CNN test	Adopted
11	Calibration closes most of the CNNs' head start: two +6.7 before, +2.4 after (CNN-2)	The CNN's main strength is for a new user without calibration, which the product does not use	Insight
12	two beats the calibrated LDA for all 19 people, most for the hardest ones (CNN-2)	Below the +3 bar, so not adopted; kept on record	Closed by the rule
13	The calibration settings chosen for (b)–(d) were at the edge of the range tried (CNN-2)	Re-check with a wider range (task 6b)	Closed: same setting chosen; verdict confirmed (CNN-2b)
14	On the 12 gestures the CNN's lead is +5.4, on the 6 groups +2.4 (CNN-2)	Re-evaluate if the final design uses many more groups	Open (future)
15	One trial does almost all the work: LDA +20.5, then +1.2 for two more trials (CNN-2c)	Keep the 1-trial calibration	Confirmed
16	The CNN with 1 trial (94.3%) beats the LDA with 3 trials (93.1%) (CNN-2c)	Not adopted (chip cost); a CNN would allow a shorter calibration on a larger chip	Open (future)
17	Calibrated in one arm position, the CNN loses 9.6 points and the LDA 12.1; the CNN's lead doubles to +4.9 (CNN-4)	Not adopted; both still need calibration across positions	Open (future)
18	With one-position calibration the LDA's best share drops from 1.0 to 0.5 (CNN-4)	Keep the blend of model C; it protects partial calibrations	Confirmed (for the LDA design)

13. Limitations to keep in mind

- Only one robustness condition (arm position) was tested for the CNN; Rest, effort, electrode shift and other days were not (CNN-3 and CNN-5 dropped).
- CNN-2 tried one calibration trial per round (the LDA study averaged all three choices); both models used the same trials, so the comparison stays fair.
- CNN-2's calibration settings were first chosen at the edge of the range tried; CNN-2b widened the range and the same setting was chosen again.
- CNN-1 has no calibration; the deciding comparison with the calibrated LDA is CNN-2.
- CNN-1 ran on two GPUs (165 runs on Colab's T4, the last 6 on a laptop RTX 3050), both in float32. The random start of each run changes results far more than the GPU does.
- FORS-EMG has 19 people recorded in one session each. A CNN may need more people than that; the result will be specific to this amount of data.
- The chip cost counts multiply-adds and memory only; memory reads, control logic, area and power were not estimated, because no CNN was adopted.
- Offline accuracy is only a proxy for what the user feels; unwanted movements per minute at rest and delay were not measured for the CNN (CNN-3 dropped).